

DevOps Project 5 - Containerizing Spring Boot Application and Docker Image Scanning

Name : Arunabha Mukhopadhyay

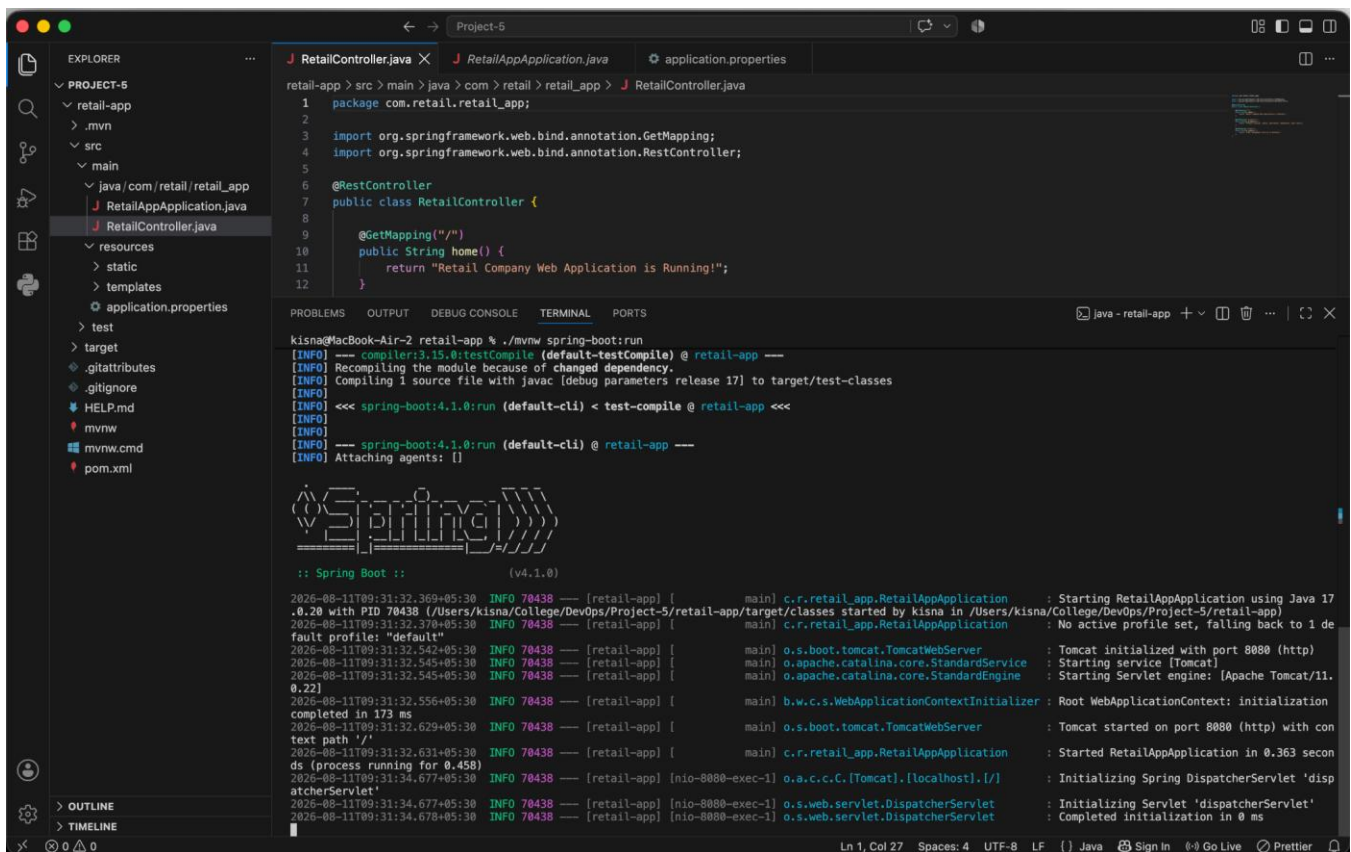
PRN.: 23070122049

Batch: 2023-27

Class : A

1. Spring Boot Application Running Successfully

This screenshot shows the Spring Boot application starting successfully on Java 17 with the embedded Tomcat server running on port 8080.



```
retail-app > src > main > java > com > retail > retail_app > J RetailController.java
1 package com.retail.retail_app;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.RestController;
5
6 @RestController
7 public class RetailController {
8
9     @GetMapping("/")
10    public String home() {
11        return "Retail Company Web Application is Running!";
12    }
13 }
```

```
2026-08-11T09:31:32.369+05:30 INFO 70438 [retail-app] [main] c.r.retail_app.RetailAppApplication : Starting RetailAppApplication using Java 17
2026-08-11T09:31:32.370+05:30 INFO 70438 [retail-app] [main] c.r.retail_app.RetailAppApplication : No active profile set, falling back to 1 de
2026-08-11T09:31:32.370+05:30 INFO 70438 [retail-app] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-08-11T09:31:32.370+05:30 INFO 70438 [retail-app] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-08-11T09:31:32.370+05:30 INFO 70438 [retail-app] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.
2026-08-11T09:31:32.370+05:30 INFO 70438 [retail-app] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization
2026-08-11T09:31:32.370+05:30 INFO 70438 [retail-app] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with con
2026-08-11T09:31:32.370+05:30 INFO 70438 [retail-app] [main] c.r.retail_app.RetailAppApplication : Started RetailAppApplication in 0.363 secon
2026-08-11T09:31:32.370+05:30 INFO 70438 [retail-app] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'disp
2026-08-11T09:31:32.370+05:30 INFO 70438 [retail-app] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-08-11T09:31:32.370+05:30 INFO 70438 [retail-app] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 0 ms
```

Figure 1: Spring Boot Application Running Successfully

2. Retail Web Application Tested in Browser

This screenshot verifies that the retail web application is accessible through the browser at localhost:8080 and returns the expected application message.

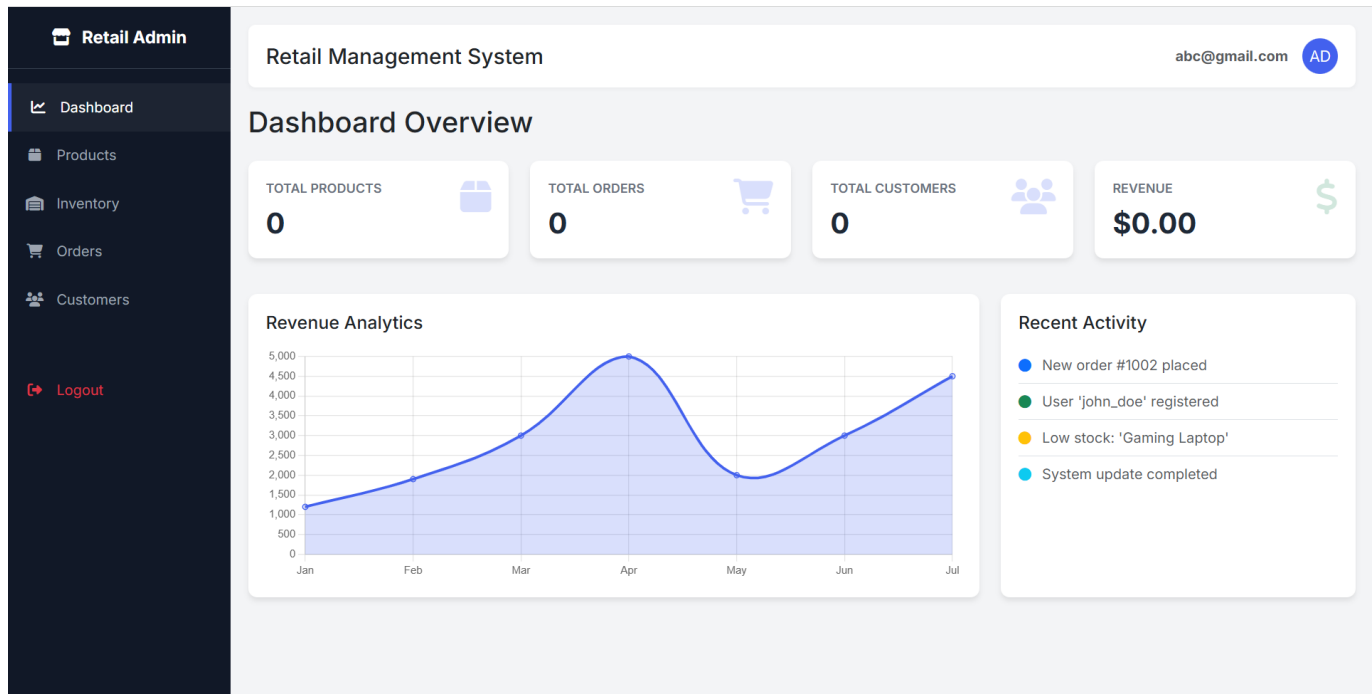


Figure 2: Retail Web Application Tested in Browser

3. Spring Boot JAR File Built Successfully

This screenshot shows Maven completing the package operation with BUILD SUCCESS and generating the retail-app-0.0.1-SNAPSHOT.jar file.

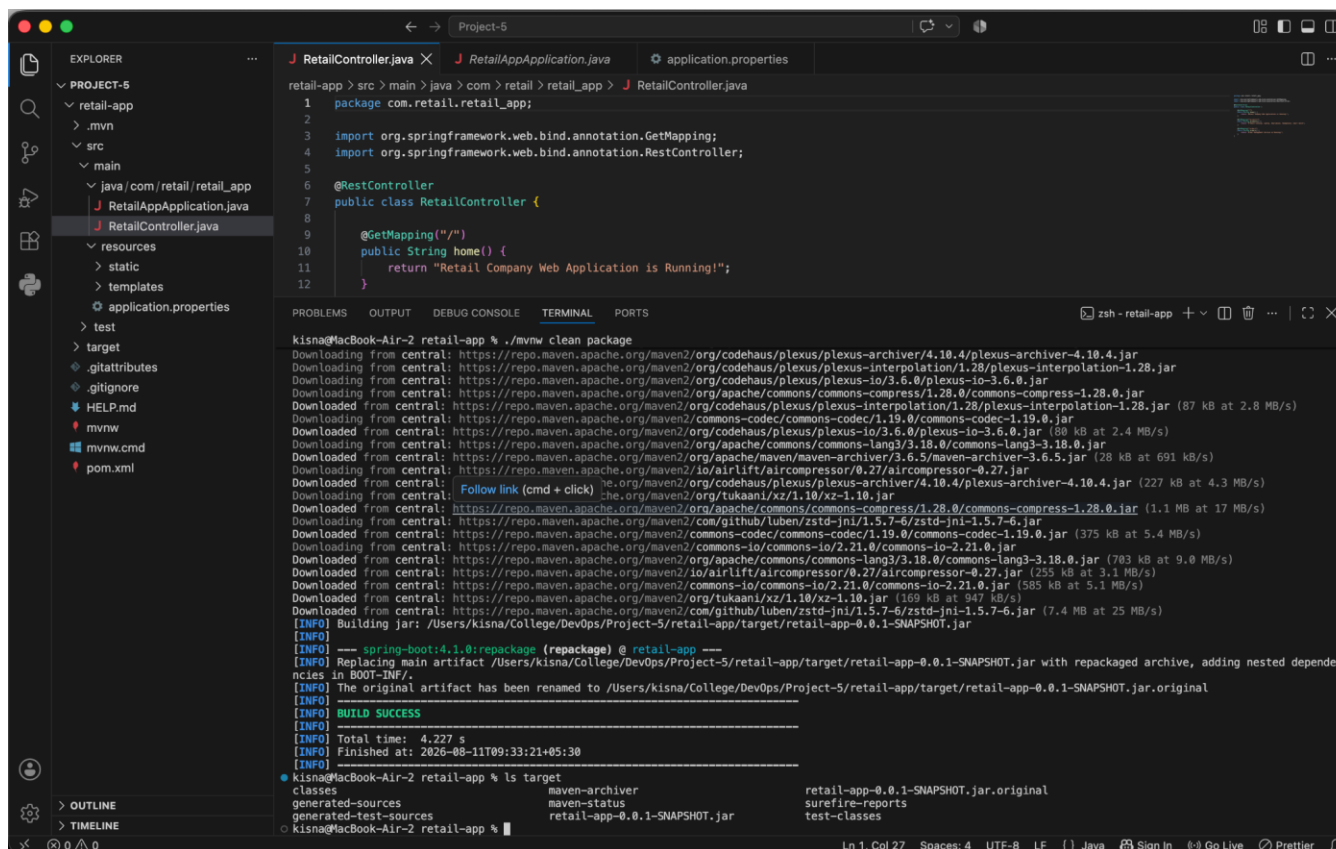
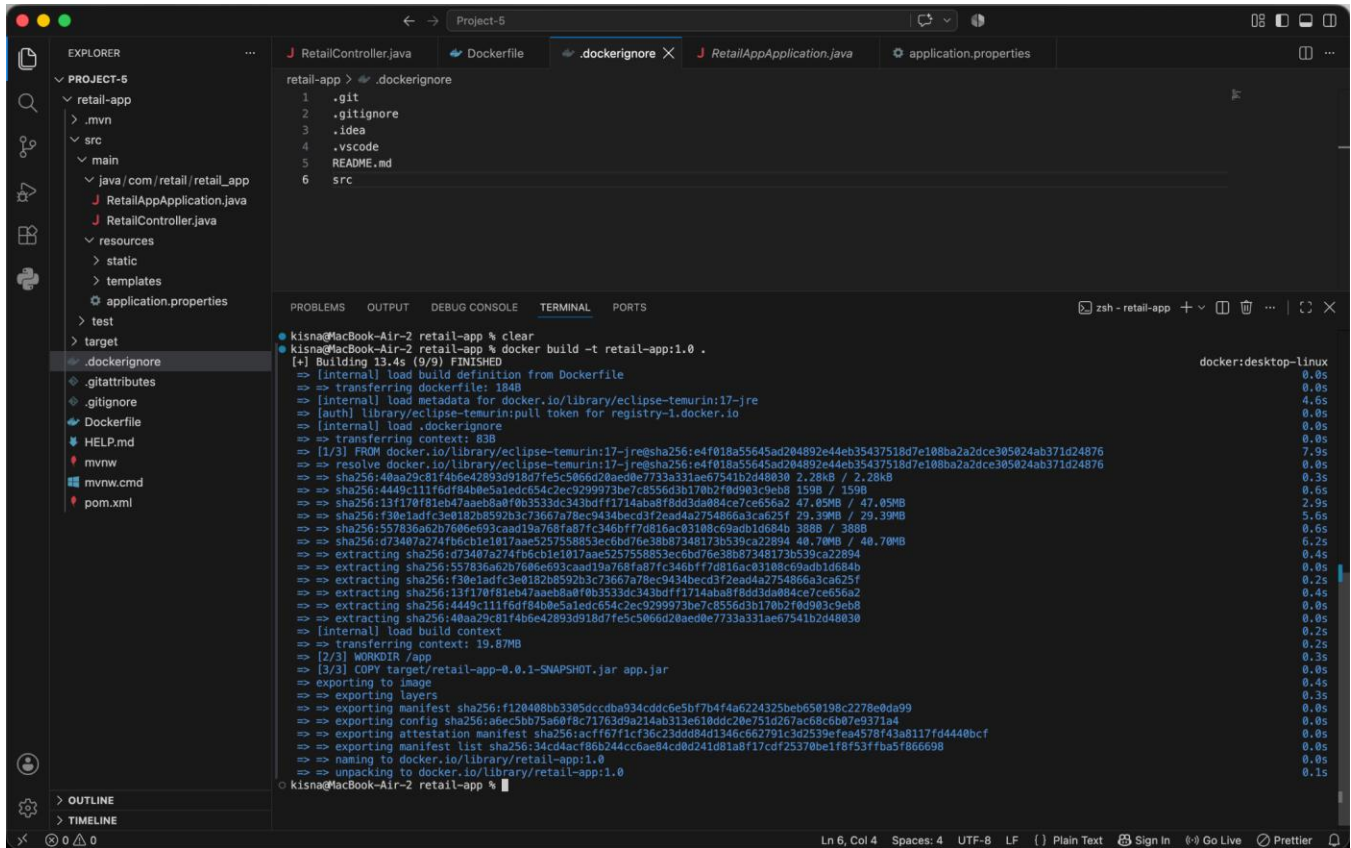


Figure 3: Spring Boot JAR File Built Successfully

4. Docker Image Successfully Built

This screenshot demonstrates the successful Docker build process using the Dockerfile and creation of the retail-app:1.0 image.



```
Project-5
J RetailController.java Dockerfile .dockerignore J RetailAppApplication.java application.properties
PROJECT-5
retail-app
  .mvn
  src
  main
    java/com/retail/retail_app
      RetailAppApplication.java
      RetailController.java
    resources
      static
      templates
    application.properties
  test
  target
  .dockerignore
  .gitattributes
  .gitignore
  Dockerfile
  HELP.md
  mvnw
  mvnw.cmd
  pom.xml
OUTLINE
TIMELINE
klsna@MacBook-Air-2 retail-app % clear
klsna@MacBook-Air-2 retail-app % docker build -t retail-app:1.0 .
[+] Building 13.4s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 184B
=> [internal] load metadata for docker.io/library/eclipse-temurin:17-jre
[auth] library/eclipse-temurin:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 83B
=> [1/3] FROM docker.io/library/eclipse-temurin:17-jre@sha256:e4f018a55645ad204892e44eb35437518d7e108ba2a2dce305024ab371d24876
=> resolve docker.io/library/eclipse-temurin:17-jre@sha256:e4f018a55645ad204892e44eb35437518d7e108ba2a2dce305024ab371d24876
=> sha256:40aa29c81f4b6e42893d918d7f6c5c66d20aed0e7733a31ae6754102d4803b 2.28kB / 2.28kB
=> sha256:4449c111f6df84b0e5a1edc654c2ec9299973be7c8556d3b170b2f8d903c9eb8 159B / 159B
=> sha256:13f170f81eb47a9eb8a0f0b3533dc343bdff1714aba8f8dd3da084ce7ce656a2 47.05MB / 47.05MB
=> sha256:f30e1adfc3e0182b8592b3c73667a78ec9434becd3f2ead4a2754866a3ca625f 29.39MB / 29.39MB
=> sha256:557836a62b7606e993caad19a768fa87fc346bfff7d816ac03108c69adb1d684b 388B / 388B
=> sha256:d73407a274f66cb1e1017aae5257558853ec6bd76e38b07340173b539ca22894 40.70MB / 40.70MB
=> extracting sha256:d73407a274f66cb1e1017aae5257558853ec6bd76e38b07340173b539ca22894 0.45s
=> extracting sha256:557836a62b7606e993caad19a768fa87fc346bfff7d816ac03108c69adb1d684b 0.0s
=> extracting sha256:f30e1adfc3e0182b8592b3c73667a78ec9434becd3f2ead4a2754866a3ca625f 0.2s
=> extracting sha256:13f170f81eb47a9eb8a0f0b3533dc343bdff1714aba8f8dd3da084ce7ce656a2 0.4s
=> extracting sha256:4449c111f6df84b0e5a1edc654c2ec9299973be7c8556d3b170b2f8d903c9eb8 0.0s
=> extracting sha256:40aa29c81f4b6e42893d918d7f6c5c66d20aed0e7733a31ae6754102d4803b 0.0s
=> [internal] load build context
=> => transferring context: 19.87MB
=> [2/3] WORKDIR /app
=> [3/3] COPY target/retail-app-0.0.1-SNAPSHOT.jar app.jar
=> exporting to image
=> exporting layers
=> exporting manifest sha256:f120408bb3305dcccba934cddc6e5b7fb4f4a6224325beb650198c2278e0da99 0.0s
=> exporting config sha256:a6ec5bb75a60f8c71763d9a214ab313e6180dc20e751d267ac68c607e9371a4 0.0s
=> exporting attestation manifest sha256:acf167f1cf36c23d0d84d1346c662791c3d2539efea4570f43a0117f64440bcf 0.0s
=> exporting manifest list sha256:34cd4ac160b244cc6aed4cd0d241081a0f17cdf25370be1f8153fba5f866698 0.0s
=> naming to docker.io/library/retail-app:1.0
=> unpacking to docker.io/library/retail-app:1.0
klsna@MacBook-Air-2 retail-app %
```

Figure 4: Docker Image Successfully Built

5. Docker Image Verified with docker images

This screenshot verifies that the retail-app:1.0 Docker image exists locally and displays its image ID and size.

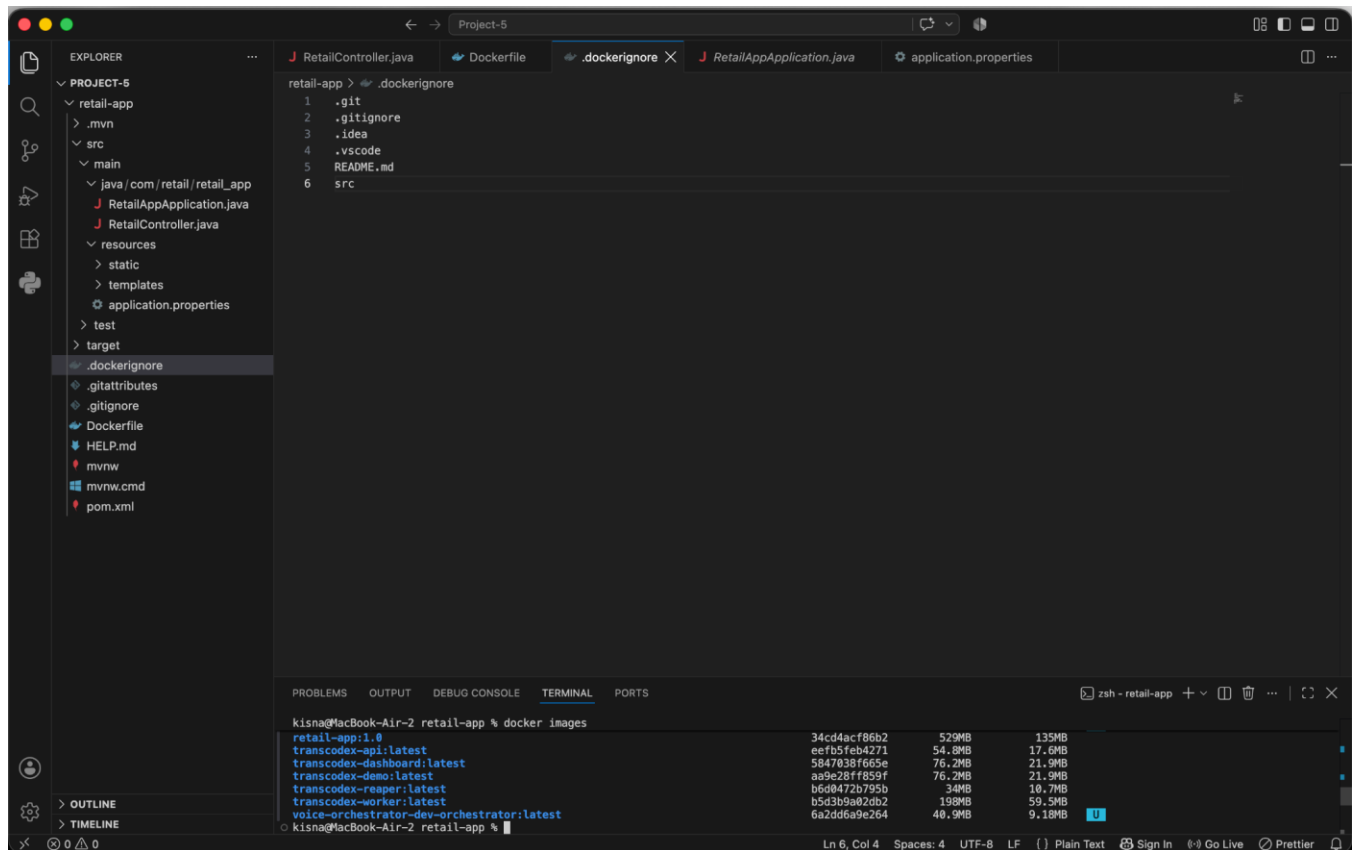


Figure 5: Docker Image Verified with docker images

6. Retail Application Container Running

This screenshot shows the retail-container running from the retail-app:1.0 image with port 8080 mapped from the host to the container.

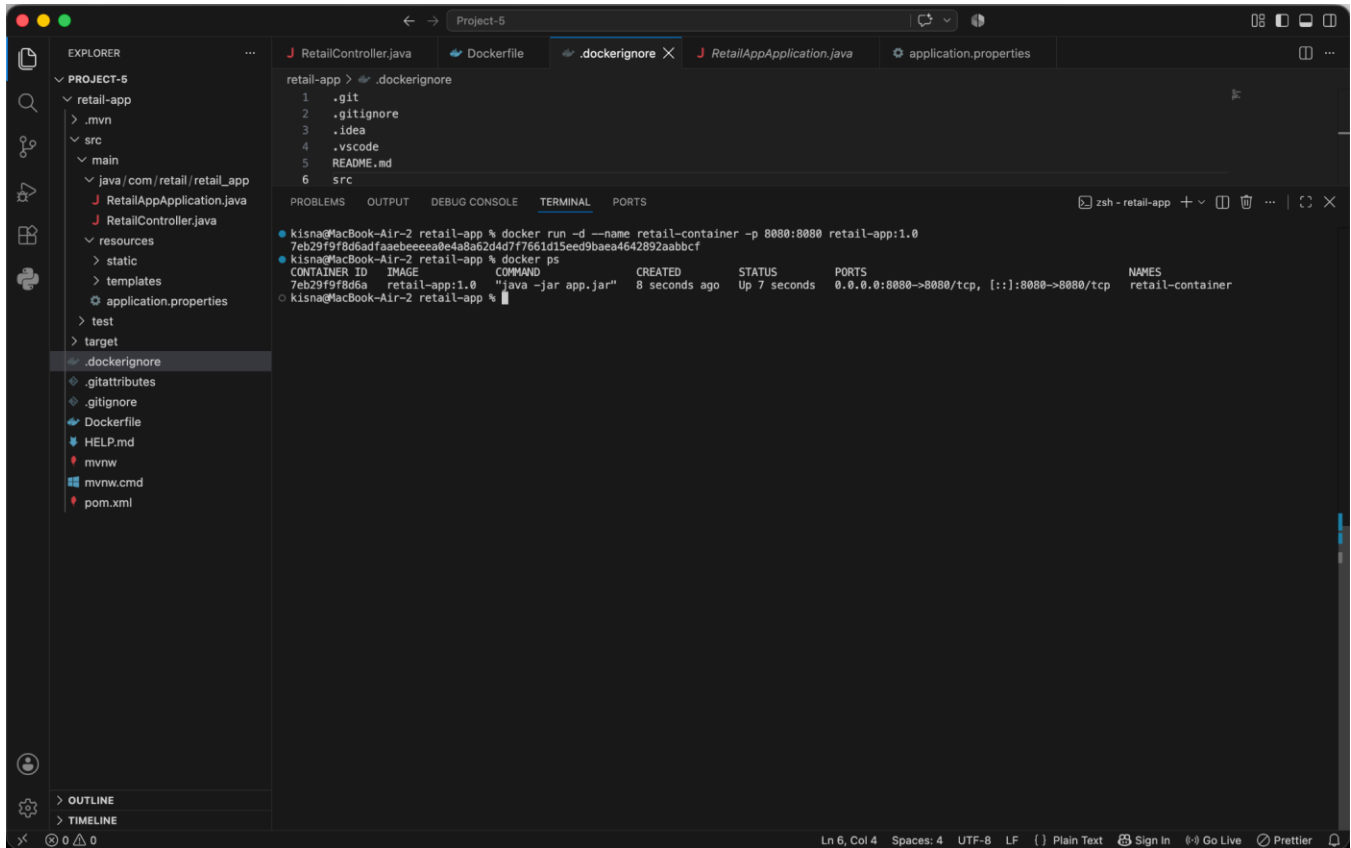


Figure 6: Retail Application Container Running

7. Dockerized Retail Application Verified in Browser

This screenshot confirms that the Spring Boot application is successfully running inside the Docker container and is accessible through localhost:8080.

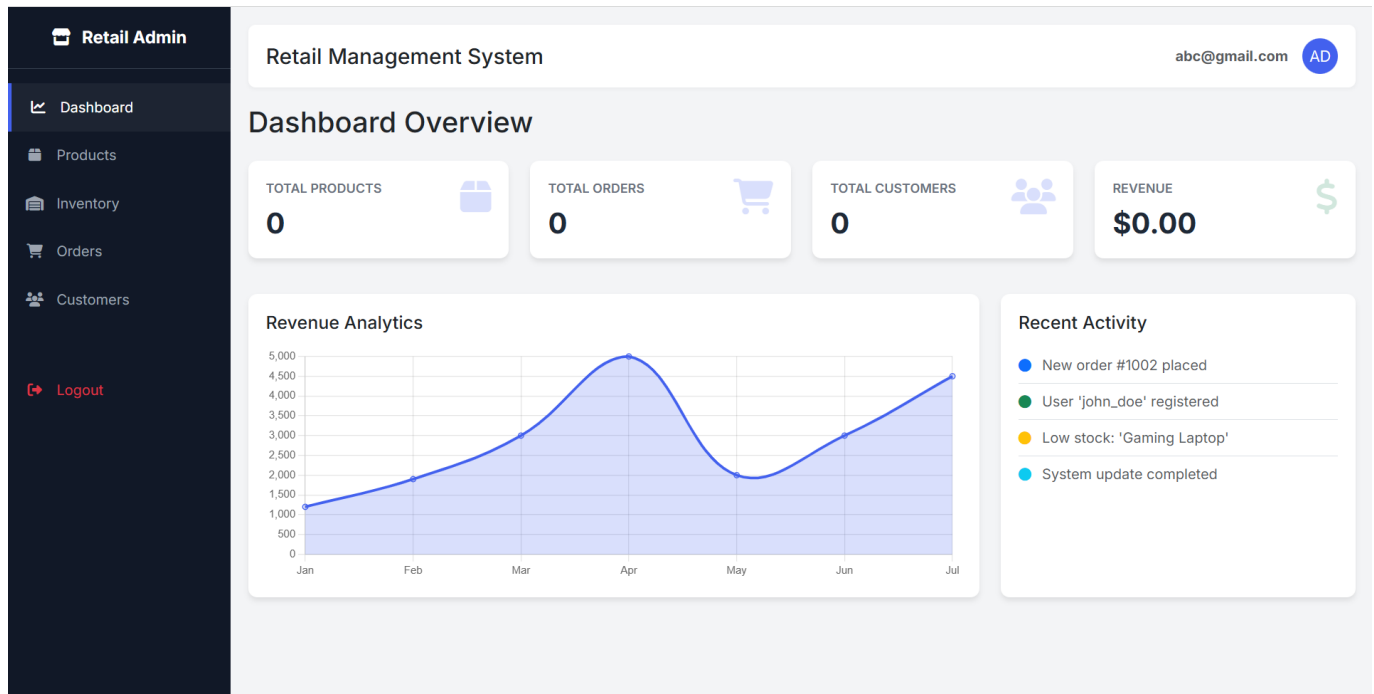
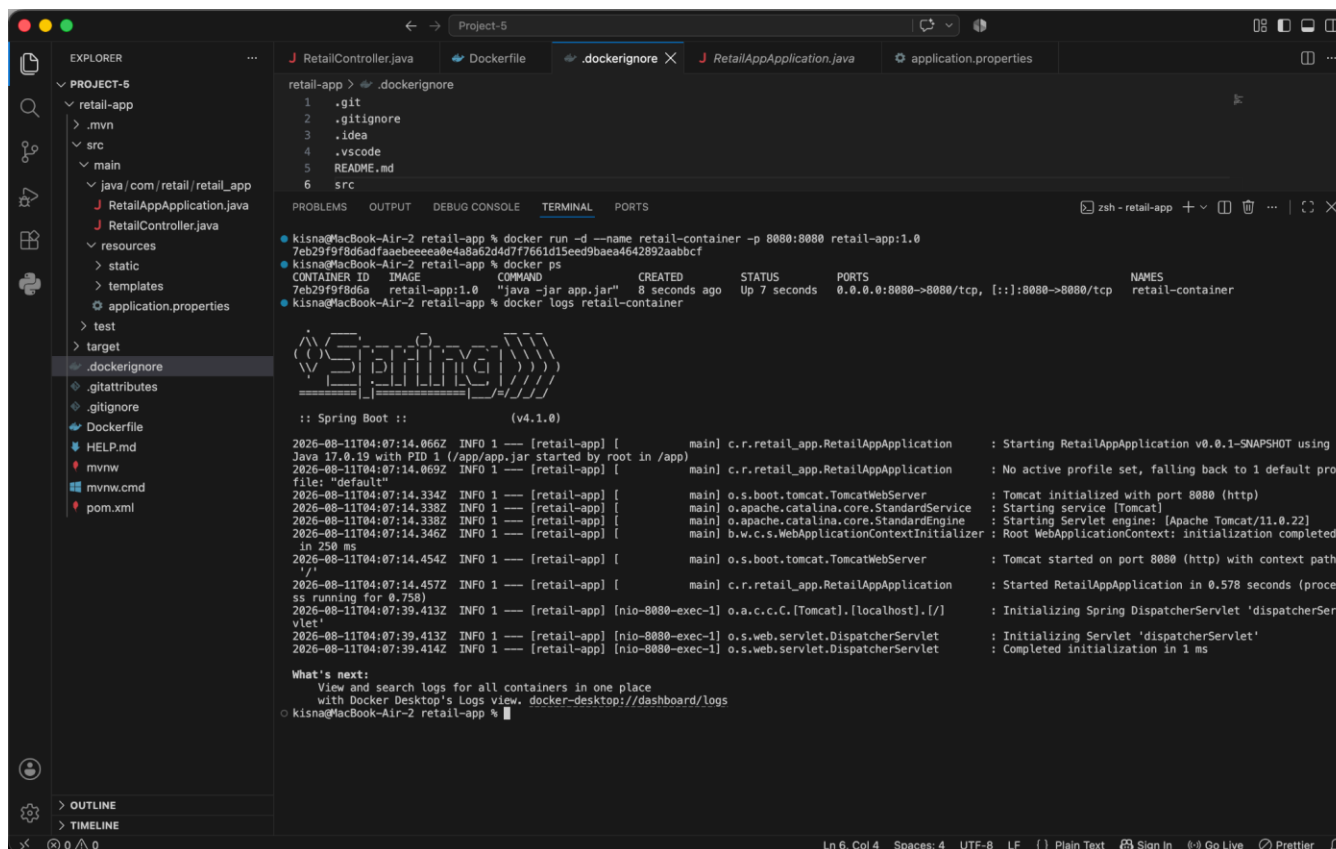


Figure 7: Dockerized Retail Application Verified in Browser

8. Docker Container Logs Showing Successful Startup

This screenshot displays the container logs, confirming that Spring Boot started successfully and Tomcat is listening on port 8080.



The screenshot shows an IDE window with the following components:

- EXPLORER:** Displays the project structure for 'Project-5', including 'retail-app' with subdirectories like 'main', 'resources', and 'test'. The 'dockerignore' file is selected.
- Terminal:** Shows the command `docker run -d --name retail-container -p 8080:8080 retail-app:1.0` and its output, including a table of container details.
- Log Output:** Displays the logs for the 'retail-container', showing the Spring Boot startup process, including the 'Spring' logo, version (v4.1.0), and the initialization of Tomcat and the application.

The log output includes the following information:

- Container ID: 7eb29f9f8d6a
- Image: retail-app:1.0
- Command: `java -jar app.jar`
- Status: Up 7 seconds
- Ports: 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
- Names: retail-container

The log output shows the following messages:

- `2026-08-11T04:07:14.066Z INFO 1 --- [retail-app] [main] c.r.retail_app.RetailAppApplication : Starting RetailAppApplication v0.0.1-SNAPSHOT using Java 17.0.19 with PID 1 (/app/app.jar started by root in /app)`
- `2026-08-11T04:07:14.069Z INFO 1 --- [retail-app] [main] c.r.retail_app.RetailAppApplication : No active profile set, falling back to 1 default profile: "default"`
- `2026-08-11T04:07:14.334Z INFO 1 --- [retail-app] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)`
- `2026-08-11T04:07:14.338Z INFO 1 --- [retail-app] [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]`
- `2026-08-11T04:07:14.338Z INFO 1 --- [retail-app] [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]`
- `2026-08-11T04:07:14.346Z INFO 1 --- [retail-app] [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 250 ms`
- `2026-08-11T04:07:14.454Z INFO 1 --- [retail-app] [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'`
- `2026-08-11T04:07:14.457Z INFO 1 --- [retail-app] [main] c.r.retail_app.RetailAppApplication : Started RetailAppApplication in 0.578 seconds (process running for 0.758)`
- `2026-08-11T04:07:39.413Z INFO 1 --- [retail-app] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'`
- `2026-08-11T04:07:39.413Z INFO 1 --- [retail-app] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'`
- `2026-08-11T04:07:39.414Z INFO 1 --- [retail-app] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms`

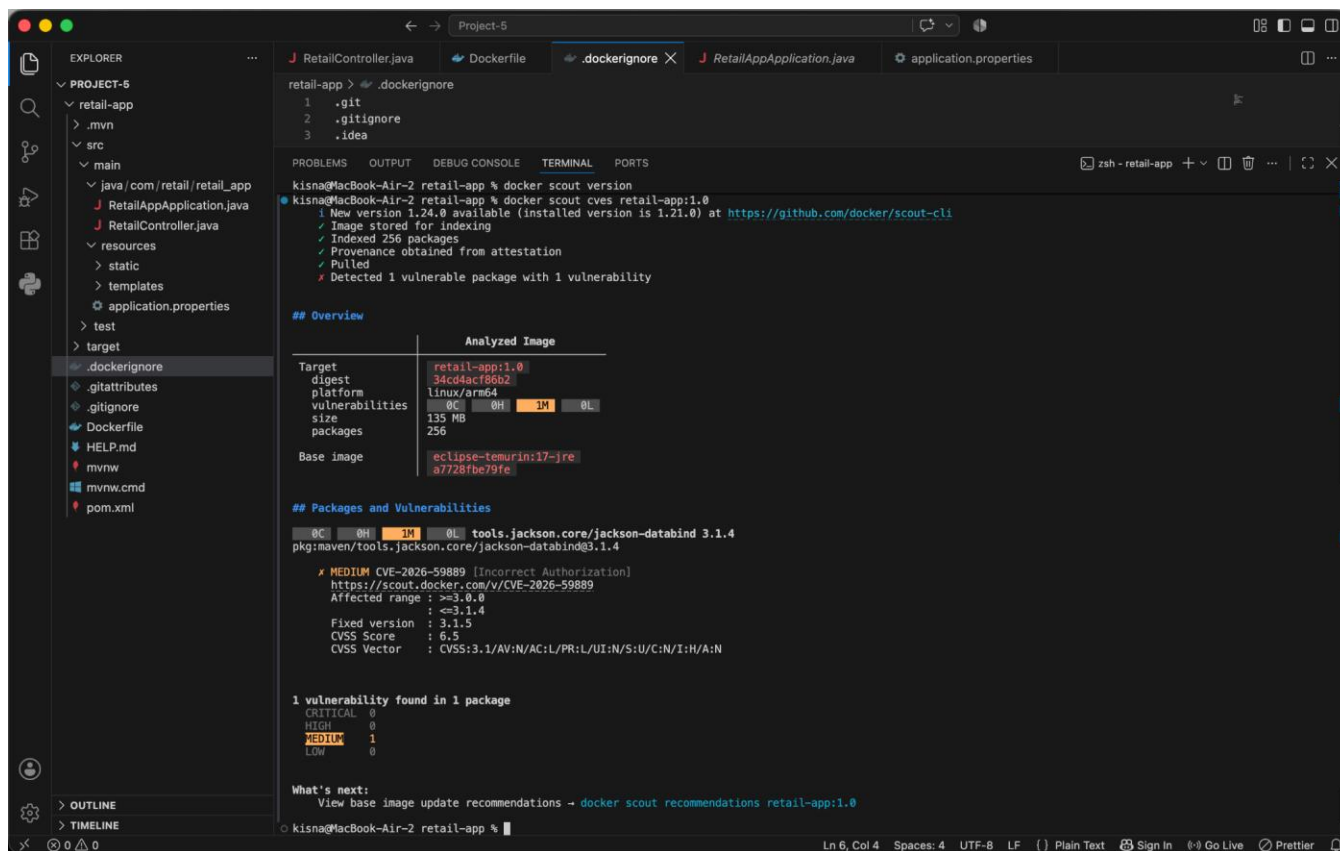
The log output also includes a 'What's next:' section with the following text:

- View and search logs for all containers in one place with Docker Desktop's Logs view. [docker-desktop://dashboard/logs](#)

Figure 8: Docker Container Logs Showing Successful Startup

9. Docker Scout Vulnerability Scan

This screenshot shows Docker Scout scanning the retail-app:1.0 image and identifying one medium-severity vulnerability in a Jackson databind package.



```
retail-app > .dockerignore
1 .git
2 .gitignore
3 .idea

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
kiswa@MacBook-Air-2 retail-app % docker scout version
kiswa@MacBook-Air-2 retail-app % docker scout cves retail-app:1.0
1 New version 1.24.0 available (installed version is 1.21.0) at https://github.com/docker/scout-cli
✓ Image stored for indexing
✓ Indexed 256 packages
✓ Provenance obtained from attestation
✓ Pulled
✗ Detected 1 vulnerable package with 1 vulnerability

## Overview

Analyzed Image
Target retail-app:1.0
digest 34cd4acf86b2
platform linux/arm64
vulnerabilities 0C 0H 1M 0L
size 135 MB
packages 256
Base image eclipse-temurin:17-jre
a7728fbc79fe

## Packages and Vulnerabilities
0C 0H 1M 0L tools.jackson.core/jackson-databind 3.1.4
pkg:maven/tools.jackson.core/jackson-databind@3.1.4

✗ MEDIUM CVE-2026-59889 [Incorrect Authorization]
https://scout.docker.com/v/CVE-2026-59889
Affected range : >=3.0.0
                 : <=3.1.4
Fixed version : 3.1.5
CVSS Score : 6.5
CVSS Vector : CVSS:3.1/AV:N/AC:L/PR:L/UI:N/S:U/C:N/I:H/A:N

1 vulnerability found in 1 package
CRITICAL 0
HIGH 0
MEDIUM 1
LOW 0

What's next:
View base image update recommendations -> docker scout recommendations retail-app:1.0
kiswa@MacBook-Air-2 retail-app %
```

Figure 9: Docker Scout Vulnerability Scan

10. Docker Image Layer History

This screenshot displays the Docker image layer history, showing the Dockerfile instructions and layers used to construct the retail-app:1.0 image.

The screenshot shows an IDE window with the following components:

- EXPLORER:** A file tree on the left showing the project structure. The file `.dockerignore` is selected.
- Terminal:** A terminal window at the bottom showing the command `docker history retail-app:1.0` and its output.
- Table:** A table displaying the Docker image layer history for the `retail-app:1.0` image.

The table data is as follows:

IMAGE	CREATED	CREATED BY	SIZE	COMMENT
34cd4acf86b2	6 minutes ago	ENTRYPOINT ["java" "-jar" "app.jar"]	0B	buildkit.dockerfile.v0
<missing>	6 minutes ago	EXPOSE [8080/tcp]	0B	buildkit.dockerfile.v0
<missing>	6 minutes ago	COPY target/retail-app-0.0.1-SNAPSHOT.jar ap...	19.9MB	buildkit.dockerfile.v0
<missing>	6 minutes ago	WORKDIR /app	8.19kB	buildkit.dockerfile.v0
<missing>	7 days ago	ENTRYPOINT ["/_cacert_entrypoint.sh"]	0B	buildkit.dockerfile.v0
<missing>	7 days ago	COPY --chmod=755 entrypoint.sh /_cacert_ent...	12.3kB	buildkit.dockerfile.v0
<missing>	7 days ago	RUN /bin/sh -c set -eux; echo "Verifying...	12.3kB	buildkit.dockerfile.v0
<missing>	7 days ago	RUN /bin/sh -c set -eux; ARCH=\$(dpkg --...	143MB	buildkit.dockerfile.v0
<missing>	7 days ago	ENV JAVA_VERSION=jdk-17.0.19+10	0B	buildkit.dockerfile.v0
<missing>	7 days ago	RUN /bin/sh -c set -eux; apt-get update;...	95.6MB	buildkit.dockerfile.v0
<missing>	7 days ago	ENV LANG=en_US.UTF-8 LANGUAGE=en_US:en LC_AL...	0B	buildkit.dockerfile.v0
<missing>	7 days ago	ENV PATH=/opt/java/openjdk/bin:/usr/local/sb...	0B	buildkit.dockerfile.v0
<missing>	7 days ago	ENV JAVA_HOME=/opt/java/openjdk	0B	buildkit.dockerfile.v0
<missing>	2 weeks ago	umoci raw add-layer --image /home/build/roc...	12.3kB	Add rock control metadata
<missing>	2 weeks ago	umoci config --image /home/build/rockcraft-	0B	Set annotations
<missing>	2 weeks ago	umoci config --image /home/build/rockcraft-	0B	Set labels
<missing>	2 weeks ago	umoci config --image /home/build/rockcraft-	0B	Set default PATH for bare-based rock
<missing>	2 weeks ago	umoci config --image /home/build/rockcraft-	0B	Set default commands
<missing>	2 weeks ago	umoci config --image /home/build/rockcraft-	0B	Set entrypoint
<missing>	2 weeks ago	umoci raw add-layer --image /home/build/roc...	135MB	

Figure 10: Docker Image Layer History